# Arpan Ghosh

Machine Learning · Data Science · AI Engineer

arpanghosh6734@gmail.com
+91 9073563551 · Kolkata, West Bengal
github.com/arpan-ghosh-42
linkedin.com/in/arpan-ghosh-21261327a

## SUMMARY

Builder at heart who loves turning real-life problems into working products across machine learning, deep learning, Generative AI, agentic AI and LLMs, now working hands-on with LLM and REST APIs. Research-minded, with a strong interest in statistics and computer vision. Seeking ML, Data Science and AI Engineer internships.

## EDUCATION

**M.Tech (CSE)**, Tripura University — 2026 – 2028 · Pursuing

**BE (CSE)**, Chandigarh University — 2021 – 2025 · CGPA 7.4

**Class XII**, WBCHSE · **Class X**, WBBSE — 93.4% · 87%

## STACK

**ML & statistics:** scikit-learn, XGBoost, SVM, regression, ensembles, clustering, PCA, SciPy

**Data & viz:** SQL, pandas, NumPy, Matplotlib, Seaborn

**LLMs & APIs:** RAG, LangChain, ChromaDB, Hugging Face, Groq & OpenAI APIs, REST APIs, Tavily, RDAP

**Deep learning:** PyTorch, TensorFlow, ANN / FNN, CNN, RNN, U-Net

**Vision & audio:** OpenCV, dlib, face & speaker embeddings, Resemblyzer, librosa

**Tools & infra:** Python, Flask, Streamlit, Git, Docker, Render, Supabase, Jupyter

## PROJECTS

### NibbleContract AI · Contract & Internship Authenticity Checker

Live demo · GitHub

Python · Flask · Groq LLM · OpenAI-compatible API · pypdf · RDAP · Tavily

- Turns internship offers, rental and loan agreements (PDF or text) into a plain-language risk report: ranked red flags, deadlines, money terms and questions to ask before signing.
- Explainable 0–100 fake-internship trust score: the LLM only extracts evidence, fixed rules score it, and unverified quotes are discarded; RDAP domain-age, SSRF-safe site checks and a web + Reddit review search (Tavily) add signals.
- Tested on 30 real documents: 90% of red flags caught, 93% fake-offer detection accuracy.

### Attendly · AI Attendance with Face & Voice Recognition

Live demo · GitHub

Python · Streamlit · dlib ResNet embeddings · scikit-learn SVC · Resemblyzer · librosa · SQL

- Marks a whole class from one group photo: 128-d face embeddings plus an SVC classifier identify each student, and a distance threshold rejects unknown faces.
- Voice roll-call matches each speaker in a classroom recording by embedding similarity; teacher and student portals with secure logins and link-based enrollment.
- 95% face-match accuracy across 25 students, 2% false-accept and 4% false-reject rate.

### Movexa · AI Gym Coach & Live Form Tracker

Live demo · GitHub

Python · Pose estimation (33 landmarks) · RAG · LLM API · Docker · WebRTC

- Builds a personalised gym or home workout plan (sets, reps, calorie and protein targets) from body stats, goal and equipment, grounded by RAG over exercise-science guidance and adapted around injuries.
- Live form coach: camera pose estimation measures joint angles, counts reps and sets, and gives on-screen and spoken cues when form breaks; 54 exercises, 95% rep-count accuracy, ~30 ms per frame.

### Machine Learning & Deep Learning Research

ML repo · DL repo

scikit-learn · XGBoost · PyTorch · TensorFlow / Keras · pandas · Matplotlib · Seaborn

- **Machine learning:** supervised and unsupervised learning on real datasets, from regression, SVM and ensembles to clustering, PCA and anomaly detection, compared through visualization.
- **Deep learning:** ANN / FNN, CNN and RNN models in PyTorch and TensorFlow for tabular, image and text tasks, plus a RAG pipeline with LangChain and ChromaDB.
- **Minor projects:** CreditWise Loan Approval · House Prices · Insurance Charges · Employee Turnover · Heart Disease · Titanic · Shop Smart · Thyroid Anomaly · Date Fruit · Power Plant · Handwritten Characters · ASL Sign Language